

NayePankh Foundation AI Assistant

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Introduction

For this project, I created an AI chatbot called NayePankh Foundation AI Assistant. The chatbot is designed to answer questions related to NayePankh Foundation such as internship opportunities, volunteer programs, and general information about the organization.

I wanted to build something practical instead of only learning theory. Since chatbots are widely used today, I thought it would be a good opportunity to learn how AI can be connected with real applications.

About the Project

The chatbot works through Telegram. A user can send a message to the bot, and the bot replies with information based on the knowledge provided to it.

I built the workflow using n8n. The chatbot uses an AI model through Groq to understand the user's questions and generate responses. I also added memory so that the bot can remember recent messages and continue the conversation more naturally.

The main purpose of the project is to provide quick information to users without requiring them to search manually for every detail.

Development Process

I started by creating a Telegram bot and connecting it with n8n. After that, I created the workflow and connected the AI Agent with the language model.

The next step was writing instructions for the chatbot so that it could answer questions specifically related to NayePankh Foundation. I tested different questions and modified the prompts until the responses became more accurate.

I also added memory functionality so that the chatbot could remember recent interactions during a conversation.

While building the project, I faced some issues with workflow configuration and Telegram integration, but after testing and debugging the workflow multiple times, I was able to make everything work correctly.

Technologies Used

The main technologies used in this project are n8n, Telegram Bot API, Groq AI, Docker, and Simple Memory.

n8n was used to create the workflow, Telegram was used as the user interface, Groq AI was used to generate responses, and Docker was used to run the project locally.

How the Chatbot Works

When a user sends a message through Telegram, the message is received by the Telegram Trigger node in n8n.

The message is then passed to the AI Agent, which processes the query and sends it to the language model. The memory component stores recent messages so that the chatbot can maintain conversation context.

After generating a response, the workflow sends the reply back to the user through Telegram.

What I Learned

This project helped me understand how AI chatbots are built and integrated with external applications.

I learned how workflow automation works in n8n, how APIs can be connected to applications, how prompts affect AI responses, and how memory can improve chatbot conversations.

I also gained practical experience in testing, debugging, and improving a real project.

Conclusion

Building the NayePankh Foundation AI Assistant was a valuable learning experience for me. It allowed me to apply concepts related to Artificial Intelligence, automation, and chatbot development in a practical way.

The chatbot is able to answer questions related to NayePankh Foundation and provide information to users through Telegram in a simple and interactive manner.

Note

This project was developed and tested locally using n8n running in Docker. Since it is not hosted on a public server, a live demo link is not available.

To demonstrate the project, screenshots, workflow files, and a video demonstration have been included with the submission.